GRE Quantitative Reasoning

모의고사 Set 7 — 문제지

최신 문제(뒷번호) · S2 고난도

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내
- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 7 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
답															

Section 1 12 Questions · 21 Minutes

Question 1 대소비교

352. Quantity A: The remainder when 2^{32} is divided by 3
Quantity B: 1

(A) Quantity A is greater.

(B) Quantity B is greater.

(C) The two quantities are equal.

(D) The relationship cannot be determined from the information given.

Question 2 대소비교

413. 1, 2, 3, ..., m
1, 2, 3, ..., n
The first sequence consists of m consecutive integers, and the second sequence consists of n consecutive integers, where m is even and n is odd.
Quantity A: The percent of integers in the first sequence that are odd
Quantity B: The percent of integers in the second sequence that are even

(A) Quantity A is greater.

(B) Quantity B is greater.

(C) The two quantities are equal.

(D) The relationship cannot be determined from the information given.

Question 3 대소비교

600. Of the 200 integers in a set, the 65th percentile is 20.
Quantity A: The arithmetic mean of the integers in the set
Quantity B: 20

(A) Quantity A is greater.

(B) Quantity B is greater.

(C) The two quantities are equal.

(D) The relationship cannot be determined from the information given.

Question 4 대소비교

778. The acreage of Hopetown, a town in Lewis County, is equal to 20 percent of the acreage of the region that is in Lewis County but not in Hopetown.
Quantity A: The acreage of Hopetown as a percent of the entire acreage of Lewis County
Quantity B: 18%

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 객관식

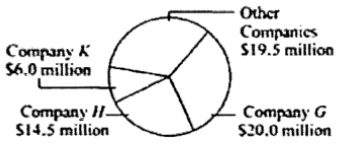
1322. If $M=(3,875)^2+(5,843)^3$, then the units digit of M is
A.2
B.4
C.7
D.8
E.9

(A) (B) (C) (D) (E)

Questions 6 to 8 are based on the following data.

Selected Data for Bookstore Companies in Region *X*

Book Sales by Bookstore Company, 1996



Year	Number of Bookstore Companies	Total Number of Bookstores in the Bookstore Companies
1992	81	260
1993	80	260
1994	75	220
1995	60	210
1996	55	215

Question 6 객관식

1264. For all bookstores in 1996, approximately what was the average (arithmetic mean) book sales per store?
A.$55,000
B.$109,000
C.$190,000
D.$280,000
E.$600,000

(A) (B) (C) (D) (E)

Question 7 객관식

1265. In 1996, companies G, H, and K owned a total of 110 bookstores. If beginning in the first half of 1997, these three companies were to purchase a total of 15 of the remaining 1996 bookstores every half year, what is the first year in which all of the 1996 bookstores would be owned by these three companies?
A.1999
B.2000
C.2003
D.2004
E.2005

A B C D E

Question 8 객관식

1266. If the total annual booksales in 1992 was $50.0 million and if this total increased by at least $1.0 million annually from 1992 to 1996, what was the greatest possible annual increase in this total from 1992 to 1996?
A.$5.0 million
B.$6.0 million
C.$7.0 million
D.$8.0 million
E.$9.0 million

A B C D E

Question 9 객관식

811. A caterer has 20 liters of fruit punch that is 12 percent grape juice, by volume. How many liters of fruit punch that is 20 percent grape juice, by volume, must be added to the original 20 liters of fruit punch in order to create fruit punch that is 15 percent grape juice, by volume?
A3
B10
C12
D15
E30

A B C D E

Question 10 주관식

444. $X=123^4-123^3+123^2-123$
What is the remainder when X is divided by 122?

Question 11 복수선택

449. At a certain company, employees who earn $20.00 per hour will be given an increase of $1.00 per hour. For each of the other employees, either the employee will be given an increase of $1.00 per hour or the employee will be given a percent increase equal to the percent increase that will be given to the employees who earn $20.00 per hour, whichever results in a larger increase for that employee. Which of the following statements are true?
Indicate all such statements.
AAn employee who earns less than $20.00 per hour will be given a percent increase that is greater than the percent increase that will be given to the employees who earn $20.00 per hour.
BAn employee who earns $22.00 per hour will be given an increase of $1.10 per hour.
CAn employee who earns $24.00 per hour will earn $25.20 per hour after the increase

Indicate all that apply. [A] [B] [C]

Question 12 객관식

789. From a group of 100 people including Alice and Bob, 40 people are to be randomly selected at the same time to win movie tickets. What is the probability that both Alice and Bob will be selected to win movie tickets?
A1,560/10,000
B1,560/9,900
C2,400/10,000
D2,400/9,900
E3,540/9,900

(A) (B) (C) (D) (E)

Section 2 15 Questions · 26 Minutes

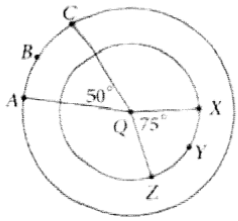
Question 1 대소비교

1178. In the xy-plane, points (-4, 0) and point (4, 0) lie on a circle C.
Quantity A: The radius of circle C
Quantity B: 4

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

1257.



Q is the center of both circles and QA=3/2(QX)
Quantity A: The length of arc ABC
Quantity B: The length of arc XYZ

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

1080. x and y are integers, and x<y−4
Quantity A: The number of even integers between x and y
Quantity B: The number of odd integers between x and y

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

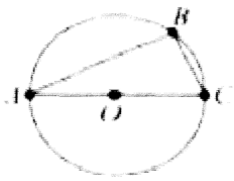
Question 4 대소비교

295. A group of n people is standing in line to purchase theater tickets. The line is a straight line with more than 10 people. Meg is the 4th person behind the person at the front of the line, and Cory is the 4th person in front of the person at the back of the line.
Quantity A: The number of people in the line between Meg and Cory
Quantity B: n−10

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 대소비교

846.

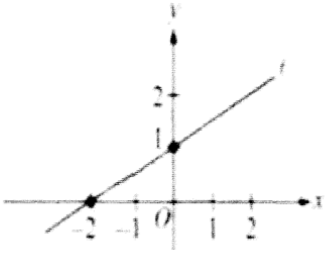


Triangle ABC is inscribed in the circle centered at O.
Quantity A: Five times the area of triangular region ABC
Quantity B: The area of the circular region

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 6 객관식

598.



In the figure above, all points below the line l satisfy which of the following inequalities?
A $y<2x-1$
B $y<2x+1$
C $y<x+1$
D $y<(x/2)-1$
E $y<(x/2)+1$

(A) (B) (C) (D) (E)

Question 7 객관식

984. Let $m=(2^3)(3^2)(5)(7^2)$ and $p=(2^2)(3^5)(5^4)(11)$. What is the greatest common divisor of m and p?
A $(2)(3)(5)$
B $(2^2)(3^2)(5)$
C $(2)(3)(5)(7)(11)$
D $(2^2)(3^2)(5)(7)(11)$
E $(2^3)(3^5)(5^4)(7^2)(11)$

A B C D E

Question 8 객관식

1112. In a certain corporation, an employee council will consist of 3 employees who will be selected from 20 eligible employees. If each of the 3 positions on the council has a different role on the council,then there are 6,840 possible 3-employee selections from the 20 eligible employees. How many possible 3-employee selections are there if the 3 positions all have the same role?
A (3)(6,840)
B (6)(6,840)
C (20)(6,840)
D 6,840/3
E 6,840/6

A B C D E

Question 9 복수선택

809. Of the 29 students in a class, 15 students study French, fewer than 16 students study Spanish, and 5 students study neither French nor Spanish. Which of the following could be the number of students in the class who study both French and Spanish? Indicate all such numbers.
A0
B3
C5
D6
E8
F9

Indicate all that apply. A B C D E F

Question 10 주관식

365. Let $1/3+1/5+1/7+1/9+1/11=m/\{(3)(5)(7)(9)(11)\}$, where m is a positive integer. What is the remainder when m is divided by 5?

Question 11 객관식

510. In a certain university, 60 percent of all sophomores are liberal arts majors, 24 percent are education majors, and the rest are majoring in other subjects or have not yet chosen a major. At the university, 55 percent of all sophomores are currently taking a psychology course. If x percent of all sophomores are liberal arts majors who are currently taking a psychology course, what is the least possible value of x?
A15
B16
C31
D36
E40

A B C D E

Question 12 객관식

710. In each round of a certain game, either 1, 3, 7, or 10 points are awarded to the winner of the round. Which of the following CANNOT be the total number of points awarded to the winner of three rounds?
A15
B16
C22
D24
E27

A B C D E

Question 13 복수선택

446. All of the 80 science students at a certain school are enrolled in at least one of three science courses: biology, chemistry, and physics. There are 60 students enrolled in biology, 50 students enrolled in chemistry, and 35 students enrolled in physics. None of the students are enrolled in all three courses. Which of the following could be the number of students enrolled in both chemistry and physics?
Indicate all such possible values.
A0
B5
C10
D15
E20
F25
G30
H35

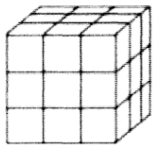
Indicate all that apply. A B C D E F G H

Question 14 객관식

875. The average (arithmetic mean) and the median of a set of 5 numbers is 0, and the set contains at least one number that is not equal to 0. Which of the following statements must be true?
AExactly one of the numbers is not equal to 0.
BExactly one of the numbers is negative.
CExactly two of the numbers are negative.
DThe set has the same number of positive numbers as negative numbers.
EThe sum of the nonzero numbers in the set is 0.

A B C D E

Question 15 객관식



467.
If 20 red cubes and 7 white cubes, all of equal size, are fitted together to form one large cube, as shown above, what is the greatest fraction of the surface area of the large cube that could be red?
A8/9
B47/54
C23/27
D5/6
E20/27

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ